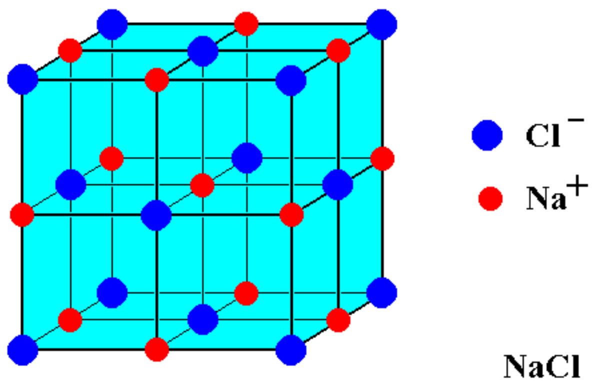
September 26, 2022

APS110 – Lecture 8

See hand written notes for drawings.

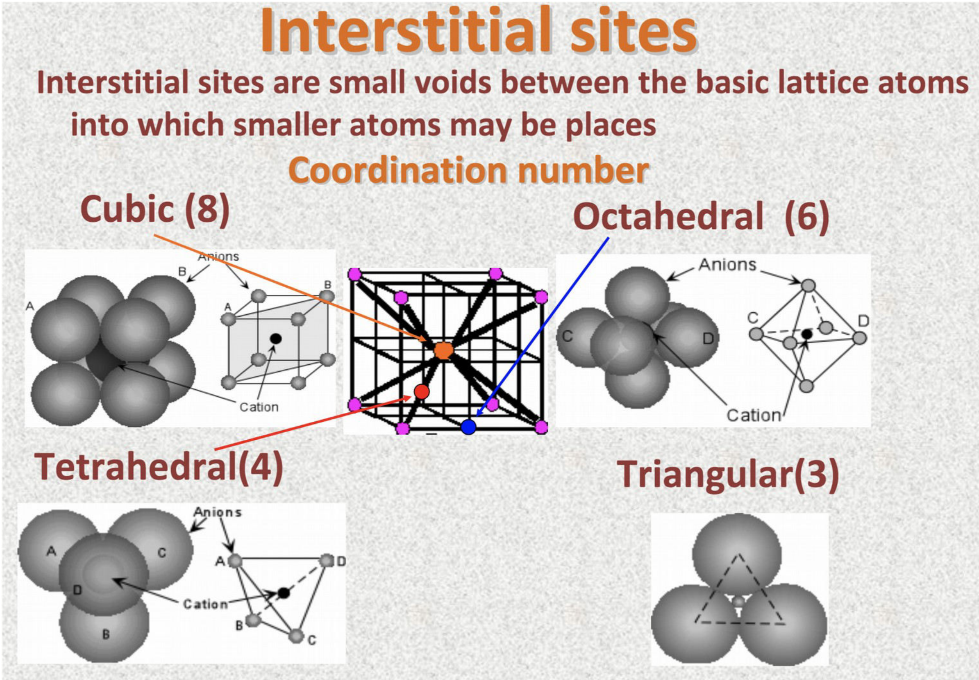
Rock Salt

- Na → cation, Cl → anion
- Cl (anion) has accepted electrons (extra electron shell), and so tends to be a little bigger
  - Because of this, we draw the anions first
- Na (cation) gets placed after, in the remaining edge positions
  - Stoiciometrically, we know there needs to be 1:1 Na:Cl atoms, and so to complete the drawing we place the last Na ion right in the **center** of the cube
  - This is a whole sphere, it's contained entirely within the cube



- ***ASIDE***: *The way we draw the structure is by convention → it doesn't actually matter if we draw the cations in the corners and the anions in the edges (the structure is equivalent)*

- The atoms in this structure are making contact with each other in DEFINED RADII
- **Interstitial site** → space between other atoms
  - There are particular interstitial sites that are of interest to us



- Density calculation
  - n_A = number of anions inside the unit cell
  - n_C = number of cations inside the unit cell
  - A_A = Molar Mass of anion
  - A_C = Molar Mass of cation
  - V_c = volume of cube
  - N_A = Avogadro's number

Ex. PbSe
- (Recall: for FCC the atoms touch across the face diagonals)
- For rock salt the atoms do not touch in the same way, even though the anion atoms appear to be in the same places
  - The only thing we know for sure is the cation contacts the nearest neighbor anions
  - The contact is along the cube edges

Another metallic crystal structure –> **Body Centered Cubic (BCC)**
- Just like with FCC, we start by placing atoms at each of the corners
- The atom in the center of the cube makes contact with those in the corners along the **cube diagonals**
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